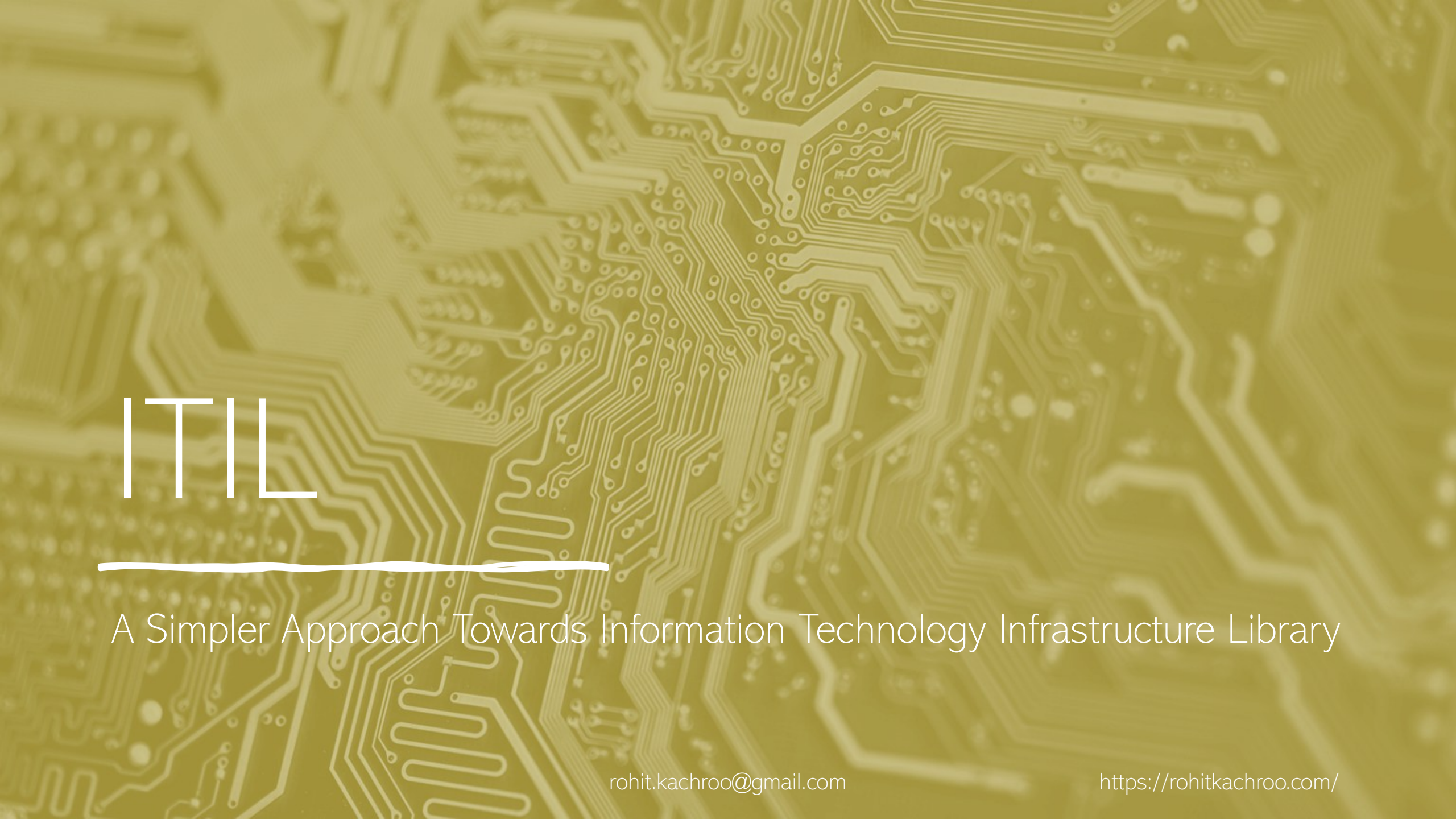




ITIL

A Simpler Approach Towards Information Technology Infrastructure Library

rohit.kachroo@gmail.com

<https://rohitkachroo.com/>

What is ITIL?

- Information technology Infrastructure library
 - Public framework describes the best practice in IT service management not a Standard
 - It focuses on the continual improvement of the quality of the IT service delivered from both business and customer perspective.
 - Benefits
 - Increased user and customer satisfaction
 - Improved service availability
 - Increased business profits and revenue
 - Financial savings from reduced rework
 - Improved decision making and optimized risk
 - Aligns IT Services with Business Objectives and plans to support the business
- Systematic approach to High Quality of IT Service Delivery
 - Provides Common Language and a guideline of expectations, roles and responsibilities

Key Concepts (1)

- VALUE Creation

- Value is created when service providers can deploy their capabilities and resources (i.e. service assets), and with a certain level of assurance, deliver to the customer a greater utility of their services.
- As established earlier, this utility is in the form of enhancing or enabling the performance of customer assets and contributing to the realization of business outcomes.'

- Utility

- What does the service do?
- Functional Requirements
- Features, inputs, outputs
- "Fit For Purpose"

- Warranty

- How well does the service do it?
- Non-functional requirements
- Capacity, performance, availability
- "Fit For Use"

- Service Owner

An individual or a team which is responsible towards customer for a given service and

- Act as prime customer contact for all service-related enquiries and issues
- Ensure that the ongoing service delivery and support meet agreed customer requirements
- Identify opportunities for service improvements
- Liaise with the appropriate process owners
- Solicit required data, statistics, and reports for analysis
- Be accountable to the IT director or Service Management director

- 4 P's of Service Management

- People – skills, training, communication
- Processes – actions, activities, changes, goals
- Products – tools, monitor, measure, improve
- Partners – specialist suppliers

- Services

- The means of delivering value to the customer by achieving the customer desired results while working within given constraints

- Service Management

- A set of specialized organizational capabilities which help us providing value to the customers in the form of services they enjoy AND/OR a set of Functions & Processes which manage the service

Key Concepts (2)

- Process

- A set of activities which are carried out in a given manner to achieve the desired objective
- A Process may define Policies, Standards, Guidelines, Activities, and Work Instructions if they are needed

- Function

- A Group of people & their tools used to carry out one or more processes or activities
- Units of organizations specialized to perform a certain type of work and responsible for specific outcomes.
- self-contained with capabilities and resources necessary for their performance and outcomes.

- A Process is

- Measurable
- Yields desired result
- Delivers result to customers
- Responds to specific events or requirements
- Closed Loop

- Staff

- The personnel who are assigned to carry out specified tasks or activities

- Roles

- A set of authorities, responsibilities & activities assigned to individual or personnel

- Metrics or KPI

- A measurement unit which imparts an ability to "Judge" the performance of a service

rohit.kachroo@gmail.com

- Process Owner

- An individual or a team which is responsible towards
- Documenting and publicizing the process
- Defining the KPIs to evaluate the effectiveness and efficiency of the process
- Reviewing KPIs and taking the required action following the analysis
- Assisting with and being ultimately responsible for the process design
- Improving the effectiveness and efficiency of the process
- Reviewing any proposed enhancements to the process
- Providing input to the ongoing Service Improvement Plan
- Addressing any issues with the running of the process
- Ensuring that all relevant staff have the required training in the process
- Ensuring that the process, roles, responsibilities, and documentation are regularly reviewed

Key Concepts (3)

- Contracts and Types

- Service Level Agreement (SLA) between customer and service provider
- Operational Level Agreement (OLA) between various internal entities and functions of an organization
- Underpinning Contract (UC) between service provider and a third party vendor for a specific customer

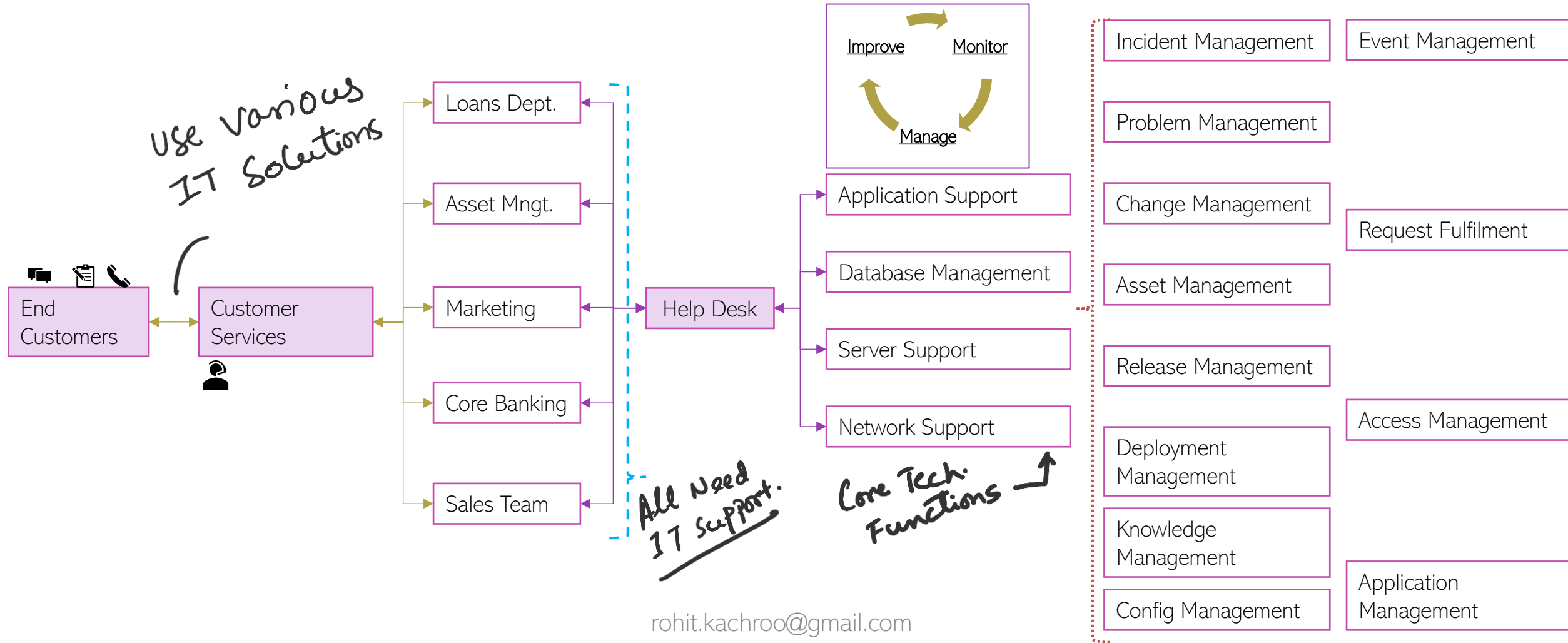
- SLA

- An agreement between a customer and a service provider defining all aspects of the service including the acceptable quality parameters on the delivered services, financial conditions and penalties.

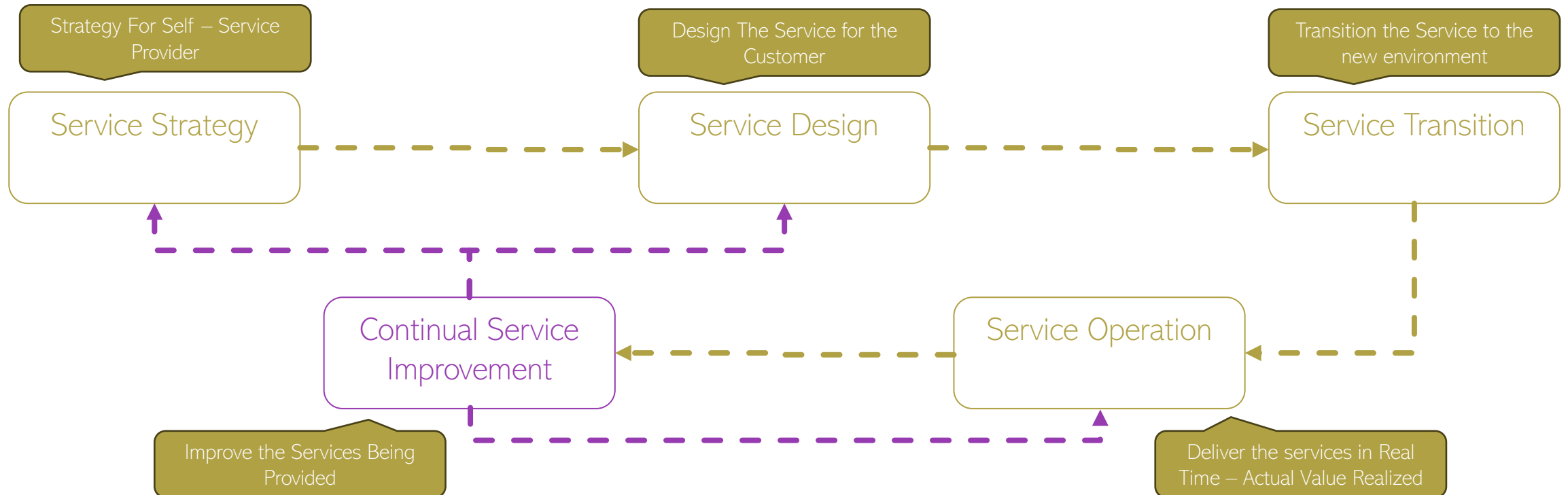
- Technical Functions

- Function which is a collection of skilled employees / leaders coming together to deliver technical capabilities

Typical Operational Scenario (e.g., Bank)



ITIL V3 Design Flow, Processes & Functions



Topics Covered Under Each Phase

SERVICE STRATEGY

- Demand Management
- Service Portfolio Management
- Financial Management

CONTINUAL SERVICE IMPROVEMENT

- 7 Step Improvement Process
- Service Reporting
- Service Measurement

SERVICE OPERATION

- Service Desk (FUNCTION)
- Incident Management
- Event Management
- Request Fulfilment
- Access Management
- Problem Management
- Technical Management (FUNCTION)
- IT Operations Management (FUNCTION)
- Application Management (FUNCTION)

SERVICE DESIGN

- Service Catalogue Management
- Service Level Management
- Availability Management
- Capacity Management
- Supplier Management
- Information Security Management
- Service Continuity Management

SERVICE TRANSITION

- Service Asset & Configuration Management
- Change Management
- Release & Deployment Management
- Knowledge Management



Service Strategy



Service Strategy

Service Strategy

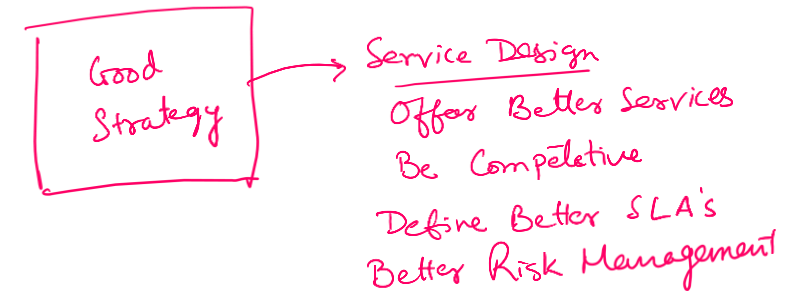
- The Service strategy volume provides guidance on how to design, develop, and implement service management not only as an organizational capability but also as a strategic asset.

Service Strategy is Effectively Managed using below topics

- Demand Management
- Financial Management
- Service Portfolio Management

Successful Service Strategy will ensure:

- IT goals are aligned to business goals.
- Annual IT initiatives that support business goals have been identified.
- There is agreement on both the strategy and a corresponding plan for achieving the goals and initiatives.
- The strategy is assessed against business outcomes.
- Opportunities for improvement are identified.



Service Strategy – Demand Management

Demand Management

- Refers to managing demands with existing customers or new customers
- Involves
 - Market research for new service development
 - Customer Requests for proposal (RFPs)
 - Works closely with sales and marketing and Portfolio Manager

Note: Demand Management can impact Service Offerings & Financial Decisions

- Customer Feedback
- Market Research
- Current Service Performance

Demand Management

Suggestions for Strategy Enhancement

Demand Management Key Factors

- If business provides excess capacity, it generates cost without creating value that provides a basis for cost recovery. Customers may be reluctant to pay for idle capacity unless has a value for them and hence is a risk to operating margins of the organizations
- Monitors Insufficient capacity as it has impact on the quality of services delivered --> Stunted growth of service

Tools to Manage Demand

- Service level agreements, forecasting, planning, and tight coordination with the customer can reduce the uncertainty in demand but cannot eliminate it.
- Techniques such as off-peak pricing, volume discounts and differentiated service levels can influence the arrival of demand in specific patterns.

Service Strategy – Portfolio Management

Portfolio Management

- Decide what services to offer
- Understand
 - Why Should a customer buy these services?
 - Why should they buy it from us?
- Provide direction to Service Design so they can manage a fully exploit the services into the future

Various Aspects / Information Sets Monitored By Portfolio Management

- Service Name
- Service Description
- Service Status
- Service Classification and criticality
- Applications used
- Data and/or data schema used
- Business owners
- IT Owners
- Service Warranty level, SLA, and SLR references
- Supporting Services and supporting resources
- Dependent Services
- Supporting OLA's Contracts and agreements
- Service Costs
- Service Charges (If Applicable)
- Service Revenue (If Applicable)
- Service Metrics

Service Strategy – Financial Management

Financial Management

- Key Questions Answered for the business decision making are:
 - Is our differentiation strategy resulting in higher profits or revenues, lower costs, or greater service adoption?
 - Which services cost us the most, and why?
 - What are our volumes and types of consumed services, and what is the correlating budget requirement?
 - How efficient are our service provisioning models in relation to alternatives?
 - Does our strategic approach to service design result in services that can be offered at a competitive 'market price', substantially reduce risk or offer superior value?
 - Where are our greatest service inefficiencies?
 - Which functional areas represent the highest priority opportunities for us to focus on as we generate a Continual Service Improvement strategy?

Service Design



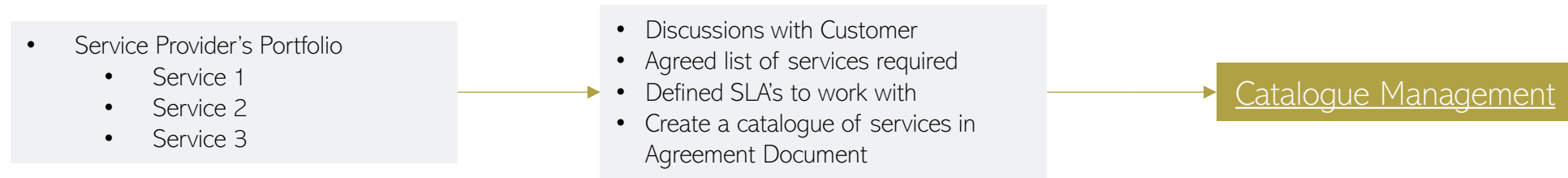
Service Design



SERVICE DESIGN

- Service Catalogue Management
 - Service Level Management
 - Availability Management
 - Capacity Management
 - Supplier Management
 - Information Security Management
 - Service Continuity Management
- Design Services for Specific Customers
 - Discussion and finalization phase
 - Services are not yet live here
 - Agreements between customer and service provider being negotiated in this phase

Service Design – Catalogue Management



High Level Concept Flow for Catalogue Management

Service Catalogue Management

- Sub Part of Portfolio Management
- List of specific Services provided
- Helps in designing manpower req.
- Also forms Basis of SLA

Service Design – Service Level Management

Service Level Management

- Set of People and systems that allow the organization to ensure that agreed quality of services are being met and that the necessary resources are being provided
- It is also responsible to plan process of
 - Negotiating
 - Defining
 - Measuring
 - Managing and
 - Improving

*All this must take place in an environment with a rapidly changing Business Needs and Technology

Responsibilities

- Identify – Customer Needs / Understand them
- Define – Service to be provided to meet needs
- Finalize – The contract (Nego. For Cost, SLA etc.)
- Monitor – Service Levels –Technical + Procedural
- Report – Drawing up Service Level Reports. Sharing with Customer and comparing with Service Level Achievements
- Review – The service together with the customer to determine opportunities for improvement.

Objectives

- Integrate elements required for the provision of IT Services
- To Create documents that clearly describes the services by the various elements
- To describe the services provided to the customer in understandable terms
- Align IT Services to Business needs
- Improve IT Service delivery in a controlled manner
- May also be the driver for Service Improvement

Some Related Documents

- Service Level Requirement (SLR)
- Service Specification Sheet
- Service Quality Plan (SQP) – KPI,CSF (How we Deliver). Covers Targets for each process
- Service Catalogue – Operational Services
- Service Level Agreement (SLA) – What we deliver. Standard for measuring IT Service
- Operational Level Agreement (OLA)
- Underpinning Contracts (UC)
- Service Level Achievement
- Service Level Reports
- Service Improvement Program (SIP)

Service Design – Availability Management

Planning & Monitoring

To achieve high level of availability, we need to

- Determine the Availability of specific services
- Designing for Availability – May be backup servers
- Designing for Recoverability
- Plan for Security Issues
- Plan Maintenance & Management
- Develop the Availability Plan for critical services

All these areas help in determining the way the services need to be delivered including SLA definition and cost estimations

Availability Management

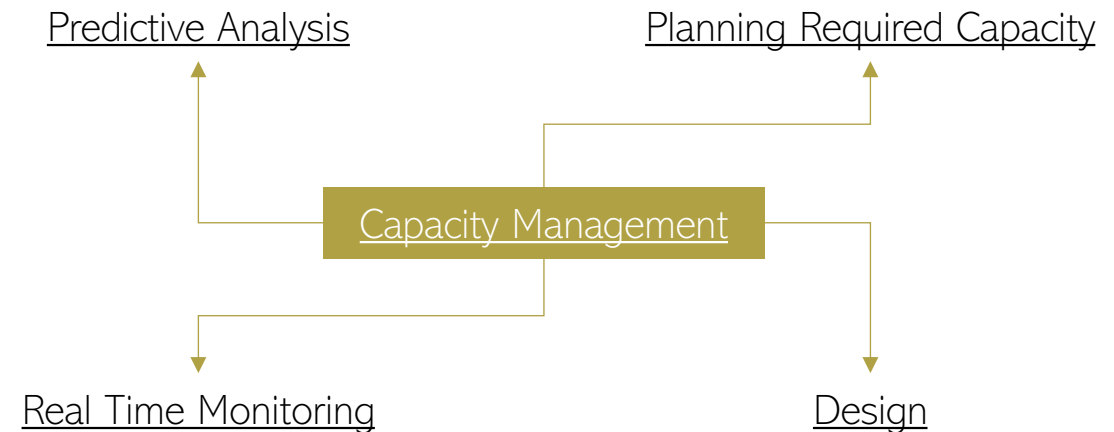
Refers to topics like

- High Availability – What are the availability criteria for the service components?
- Reliability – How reliable the infrastructure is?
- Resilience – How much tolerance does the infrastructure have in case one component fails for the service?
- Maintainability / Recoverability – How easy is it to recover the services in case of an Incident?
- Serviceability – How Easy is it to deliver support activities to the infrastructure?

Service Design – Capacity Management

Capacity Management

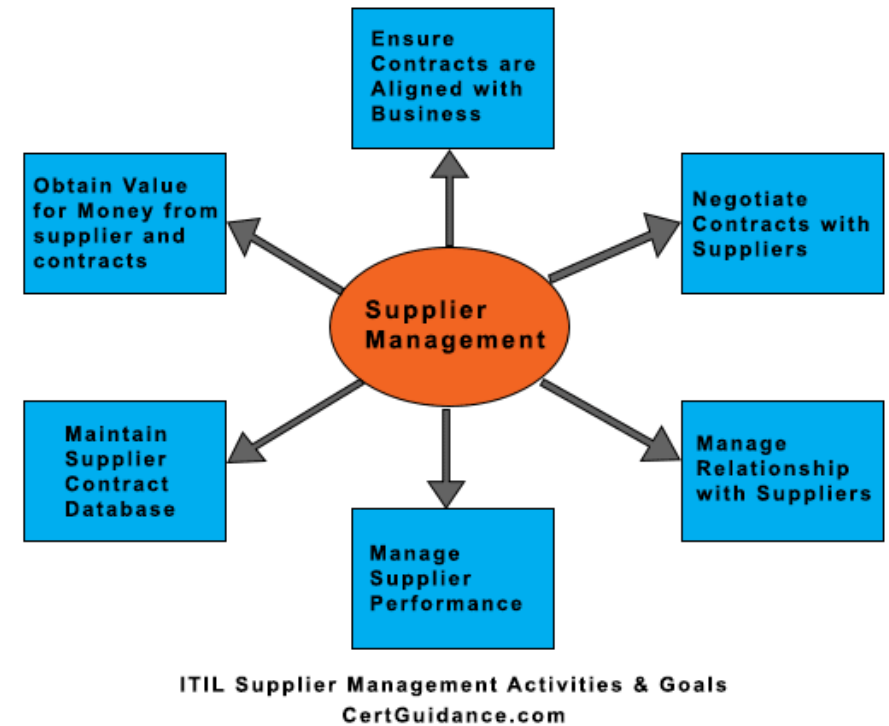
- Definition: Right Capacity, Right Time, Right Cost!
- Design your capacity requirements such that it is cost effective for current needs and is also able to fulfil future needs
- Balances Cost against Capacity so minimizes costs while maintaining quality of service



Service Design – Supplier Management

Supplier Management

- Supplier Management is where the complete relationship with vendor is Maintained
- This includes but is not limited to:
 - Monitoring Performance of the Vendor
 - Monitoring Contract and Expiry
 - Logging calls with Vendor (if req.)
 - Implement and enforce the supplier policy.
 - Management of sub-contracted suppliers.
 - Contracts categorization, selection, and risk assessment.
 - Manage & Maintain a supplier policy and a supporting Supplier and Contract Management Information System (SCMIS).
 - Maintenance of standard contracts, terms and conditions.
 - Development, negotiation, and agreement of contracts with suppliers and manage them through their lifecycle.
 - Contract review, renewal, and termination.
 - Management of suppliers and Supplier performance.
 - Managing the relationship with suppliers.
 - Management of contractual dispute resolution.
 - Agreement and implementation of service and supplier improvement plans.



Service Design – Info. Sec. Management

Information Security Management

- Deals with defining the source for the information security policies by adhering to
- Confidentiality Policies
 - Making sure only those authorized can see the data
- Integrity
 - Making sure the data is accurate and not corrupted
- Availability
 - Making sure data is supplied when it is requested



Service Design – Service Cont. Management



IT Service Continuity Management (ITSCM)

- ITSCM supports overall Business Continuity Management (BCM) by ensuring that required IT Services can be restored within agreed business timescales.
- Ensures resumption of services within agreed time-scale
- Business Impact Analysis (BIA) informs decisions about resources
- ITSCM Supports overall BCM by
 - Ensuring that required IT Infra and IT services can be restored within agreed business time-scales

Common Terms Used

- Service Continuity: A set of clearly defined plans related to how an organization will recover from a disaster and return to a pre-disaster condition, considering the four dimensions of service management.
- Business impact analysis: A key activity in the practice of service continuity management that identifies vital business functions (VBFs) and their dependencies. These dependencies may include suppliers, people, other business processes, and IT services. Business impact analysis defines the recovery requirements for IT services.

Service Transition



Service Transition



SERVICE TRANSITION

- Service Asset & Configuration Management
- Change Management
- Release & Deployment Management
- Knowledge Management

A Good Service Transition Ensures

- Set customer expectations
- Enable release integration
- Reduce performance variation
- Document and reduce known errors
- Minimize risk
- Ensure proper use of services

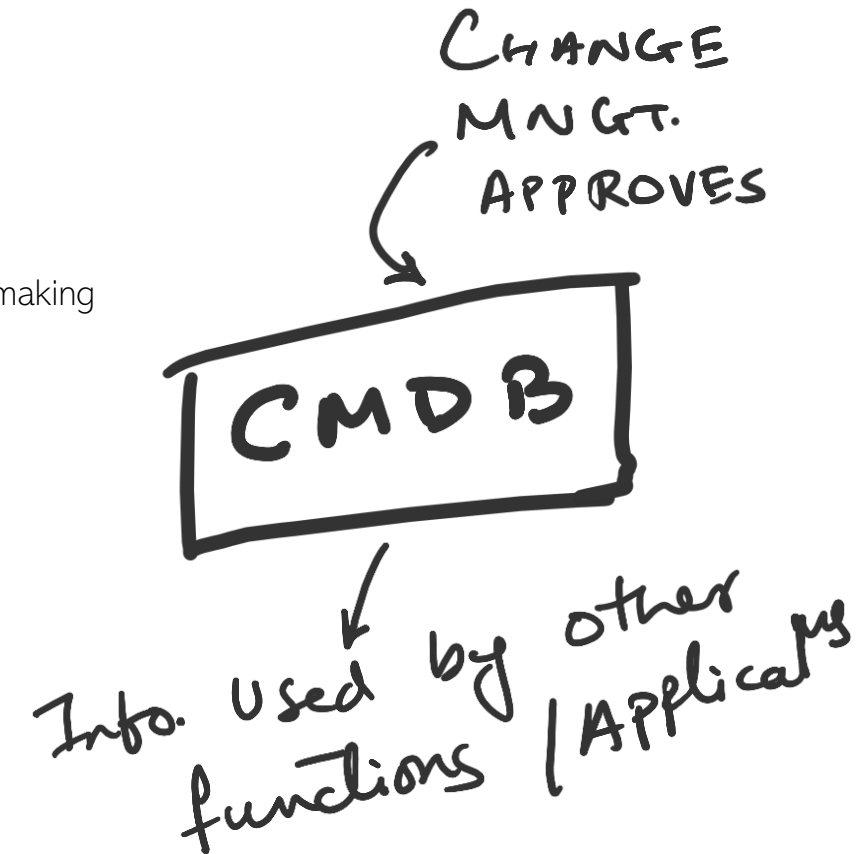
Service Asset & Configuration Management

Service Asset – What Does it Mean

- Managing Assets properly is key to successful business operations and forecasting
- Provides Logical Model of Infrastructure and Accurate Configuration information
- Controls assets and minimized costs for procurement
- Enables proper change and release management by providing accurate current asset information
- Speeds incident and problem resolution by supplying correct asset information and faster decision making

Configuration Management – What Does it Mean

- Provides reliable and up-to date information about IT Infrastructure
- Checks if changes in IT Infrastructure have been recorded correctly
- Maintains relationship between all items in IT Infrastructure
- Monitors status of all IT Components
- Ensures an accurate picture of the versions of all IT Components
- Configuration Item (CI) 
 - Any IT Component and / OR services provided by these components
- Configuration management Database (CMDB) 
 - A database with all information about the infrastructure, assets and their attributes and their relationships



Change Management

DEFINITION – CHANGE MANAGEMENT

- A Process that ensures that all changes are
 - Assessed
 - Approved
 - Implemented
 - Reviewed

In a controlled manner

DEFINITION - CHANGE

- Any deviation in the Current configuration of the IT Infrastructure is a change.

ROLES

- Change Manager -is responsible for Filtering, Accepting, Classifying, all RFC's
- Change Coordinator – Supports Change Manager in all their responsibilities

CHANGE PRIORATIZATION

Changes are assessed / Prioritized based on:

- Business Requirements
- Financial Impact
- Technical Feasibility
- Regulatory Requirements and Constraints

WHY DO WE NEED CHANGE MANAGEMENT

- Keep a track on any change in the Infrastructure
- Filtering RFC and accepting them
- Classifying the changes by category /Priority
- Consolidating, planning and coordinating the changes
- Making sure all resources are available for the change to take place.
- Involving CAB whenever necessary.
- Coordinating for closure
- Evaluating the change, if it has achieved the purpose by means of a Post Implementation Review (PIR)
- Closing the changes after review.
- Arrange all authorizations required to implement a change.

CAB MEETINGS & ATTENDEES

- Change Advisory Board - CAB
- Change Manager – Chair's the CAB

WHO ELSE ATTENDS

- Service Level manager
- Service Desk Rep / Problem Management
- IT Line Managers
- Business Managers – Client Side
- User Group Representatives
- Other Stake Holders (Requester / Supplier Representatives)

TYPES OF CHANGES

- Standard Change
 - Routine Management Tasks.
 - Handled by Service Requests
 - Example: Upgrade of Low Impact Servers, Patch update of servers with no / low impact on services
- Normal Change
 - Any change following a normal procedure
 - Gets represented as a Normal change in CAB meetings
- Emergency Change
 - Do not follow Normal CAB Procedure
 - Offline Approvals are sought
 - Triggered by Sev.1 Incidents

Release & Deployment & Knowledge Management

Release Types

Full Release

Delta / Partial Release

Package Release

Windows 11

Windows Updates / Patches

Windows Service Pack

RELEASE MANAGEMENT

- Release is a collection of authorized and tested changes ready for deployment
- A roll-out introduces a release into the live environment
- Plans and Authorizes the releases to be done in the live infrastructure

Rollout / Deployment Types

Phased Rollout

Big Bang

Iterative / Step By Step

Less Risk of Failure

Has learning Cycle

In One Go

Higher Failure Risk

KNOWLEDGE MANAGEMENT

- Vital to enabling the right information
- To be provided at the right place and the right time to the right person to enable informed decision making
- Stops data being locked away with individuals

Service Operations



Service Operations

SERVICE OPERATION

- Service Desk (FUNCTION)
 - Incident Management
 - Event Management
 - Request Fulfilment
 - Access Management
 - Problem Management
 - Technical Management (FUNCTION)
 - IT Operations Management (FUNCTION)
 - Application Management (FUNCTION)
- Maintenance
 - Management
 - Realizes Strategic Objectives
 - Is where the Value is seen and realized

Functions in the Service Operations Phase



SKILLS FOR OPERATORS

- Customer Focus
- Articulate
- Interpersonal Skills (patient!)
- Understand Business
- Methodical/Analytical
- Technical knowledge
- Multi-lingual

HELPDESK / CUSTOMER SERVICE

- 3 Major Types
 - Virtual
 - Central
 - Local
- Based on Support Strategy
- Gateway for the customer
- First point of contact
- Can be Technical / Triage / Support
- Gatekeepers of Company Information
- Coordination with Backend Teams

• Technical Management (FUNCTION)

- Comprises of the technical skills who are actually following all the defined processes.
- This function remains the backbone of service delivery as they are the people who actually troubleshoot the Incidents / Manage Changes / Solve Problems
- Helpdesk can also be a part of the technical management based on the business plan / layout

• IT Operations Management (FUNCTION)

- This is the governance layer who tracks and owns the performance of the team, plans back to green plans along with the technical management function and is responsible for the overall service delivery

• Application Management (FUNCTION)

- Is the team that manages the business critical applications, which are required to deliver the services effectively.
- Responsibility and ownership of applications and their support depends on the ownership defined in the contract / agreement (Supplier or the Customer)

Incident Management

DEALS WITH

- Unplanned interruptions to IT Services or reductions in their quality is defined as an Incident.
- End goal is to get the services back online ASAP
- Failure of a configuration item that has not impacted a service is also an incident (e.g. Disk in RAID failure)

Some KPI's Used

- Response Time
- Resolution Time
- Mean Time To Repair (MTTR)
- Mean Time Between System Failures (MTBSF)

Some Commonly Used References

- Downtime
- Up Time
- Severity
- Priority
- Urgency
- Business Impact
- Bridge Calls

Can be logged by, Any User,
Technical Staff, Monitoring Tool

Service

Technical Team
Monitoring
Services

Services Not
Available

Incident Ticket
Logged

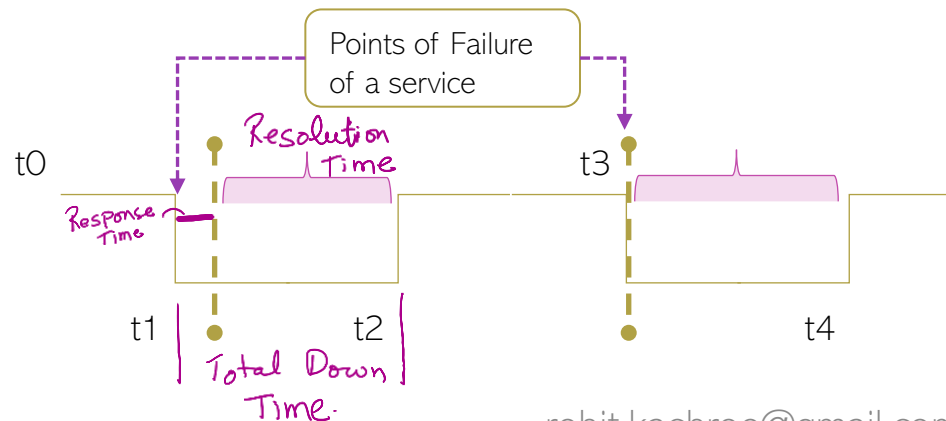
Incident
Response
Team Invoked

Bridge Calls
Arranged and
Managed by
INM Team

Constant
updates to
Business

Close Ticket
once Incident is
Resolved

Service Lifeline



- Indicates Stability Of Infrastructure
- Higher MTBSF means - More Frequent Incidents means - Lesser Stability of the Environment
- Higher MTTR Indicates Bad Support as it takes more time for the team to recover from an Incident

Problem Management

Definition

Unknown underlying cause or potential cause for one or more incidents

High Level Approach To Problem Management

- Analysis of recurring incidents
- Identify a pattern
- Conduct root cause analysis
- Suggest permanent solutions for the recurring issues
- Document and Present findings to stakeholders
- Update Known Error Database (KEDB)

Types of Problem Management Approaches

- Reactive
- Proactive

Techniques For Root Cause Analysis

- 5 Why Method
- Fish Bone Analysis

Roles

- Problem Manager
- Problem Coordinator

What Does Problem Management Deal With?

- Aims to prevent problems and resulting incidents
- Minimizes impact of unavoidable incidents
- Eliminates recurring incidents
- Proactive Problem Management
- Identifies areas of potential weakness
- Identifies workarounds
- Reactive Problem Management
- Identifies underlying causes of incidents
- Identifies changes to prevent recurrence

Multiple Incidents

Create Trend Analysis

Conduct Root Cause Analysis

Prepare RCA Document

Publish RCA & Update KEDB

Implement Solution if Approved and Authorized

Monitor Implemented Solution

Close Problem Ticket

Event Management

DEALS WITH

- Any detectable occurrence in the infrastructure that has significance for the management of the IT Infrastructure

OR

- Any change in the status of any CI in the infrastructure is defined as an Event.
- Event Management monitors all events that occur throughout the IT infrastructure
- Its purpose is to monitor normal operations and to detect and escalate exception conditions

What To Do With Events

- Need to investigate events
- Have appropriate control actions planned & documented

Types of Events

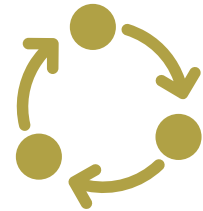
- Information
- Warning
- Exception (Higher Latency)



Request Fulfilment & Access Management

REQUEST FULFILMENT DEALS WITH

- Request Fulfilment is the process for dealing with Service Requests
- Many of them actually smaller, lower-risk, changes
- Initially managed via the Service Desk, but different organizations have different methods to handle this process now
- E.g. Employee onboarding which is a proven, repeatable, pre-approved, procedural process.



ACCESS MANAGEMENT DEALS WITH

- Access Management is the process of granting authorized users the right to use a service, while restricting access to non authorized users.
- It is based on being able accurately to identify authorized users and then manage their ability to access services as required during different stages of their Human Resources (HR) or contractual life-cycle.
- Access Management has also been called Identity or Rights Management in some organizations.



Continual Service Improvement

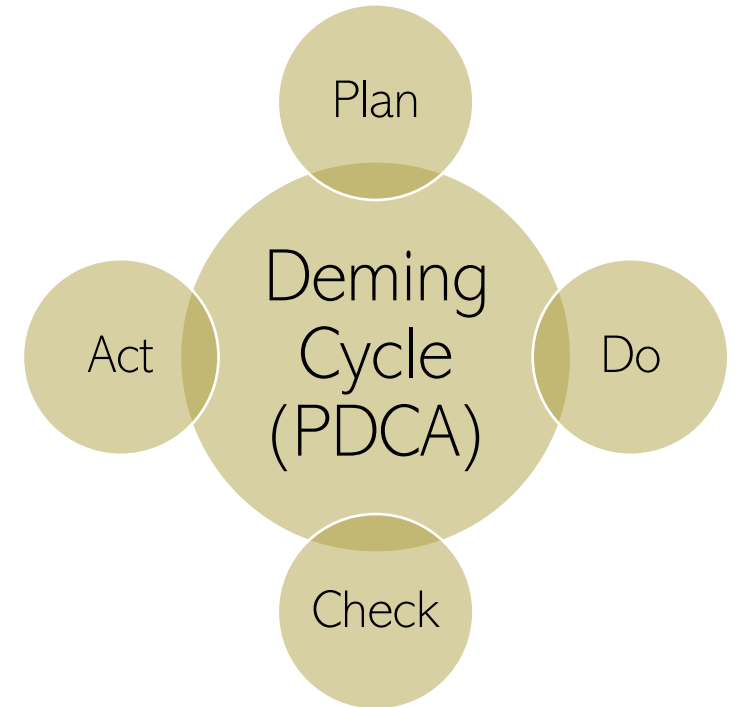
Continual Service Improvement

Continual Service Improvement

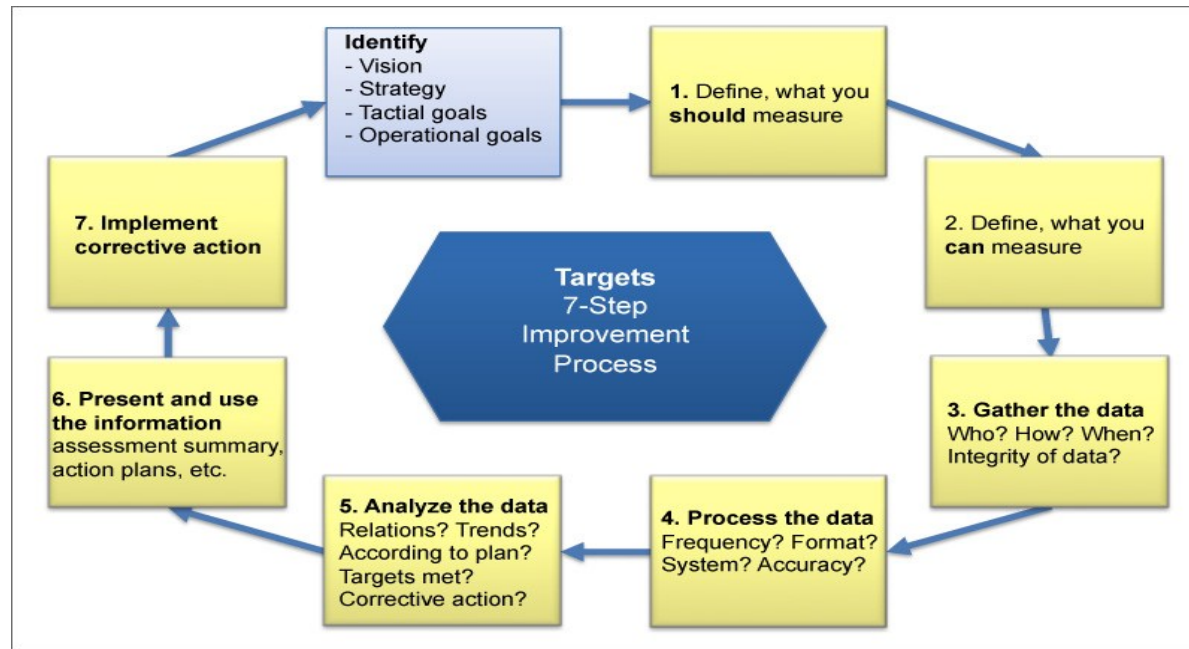
- Service Reporting
- 7 Step Improvement Plan

FOCUS FOR CSI

- Focus on Process owners and Service Owners
- Ensures that service management processes continue to support the business
- Monitor and enhance Service Level Achievements



Continual Service Improvement (CSI)



Systematic Approach Towards Service Improvement

- Works like the Deming Cycle but a bit more detailed in phases
- As an output gives improvement suggestions and motivates to implement the improvement
- Tracks and records all improvement ideas and manages organizational changes / transformations.

Service Reporting

An ideal approach to the implementation of a business-focused service reporting is to take the time to define and agree the guidelines and rules with the business and the service design. This includes:

- The target groups and their business views.
- Agreement on what is to be measured and to whom the report is to be sent.
- Basis for all calculations.
- Reporting times.
- Access to the reports and the delivery medium.
- Review meetings for improving the service report
- Also act as an input to CSI initiatives

Rough Sketches



Change Process Flow

Inc. Mngt.
Prob. Mngt
Govt. Policy
Cust. Priority

INVOKE
CHANGE
PROCESS

Once Change is Completed,
it has to be monitored.

Successful ☐ OR ☐ Unsuccessful

↓
UPDATE
CMDB
|
CLOSE CHANGE

↓
CHANGE
ROLL
BACK